

Football(Soccer) Analytics: A Case Study on the Availability and Limitations of Data for Football Analytics Research

Raheela Asif

Computer Science & Software Engineering
N.E.D University of Engineering & Technology.
Karachi, Pakistan
engr_raheela@yahoo.com

Muhammad Taha Zaheer

Computer Science & Software Engineering
N.E.D University of Engineering & Technology.
Karachi, Pakistan
taha.zaheer27@gmail.com

Saba Izhar Haque

Computer Science & Software Engineering
N.E.D University of Engineering & Technology.
Karachi, Pakistan
saba@neduet.edu.pk

Muhammad Ammar Hasan Computer Science & Software
Engineering
N.E.D University of Engineering & Technology.
Karachi, Pakistan
hasanammar94@gmail.com

Abstract— Football Analytics is fascinating area. This Case Study presents a unique situation where by a rating system for quantitatively measuring a player's performance was desired. This would eventually enable predictions derivations on various factors, such as player performance or match outcomes. Data for player rating had to be gathered from different sources; however, this Case Study outlines the solutions that were used to gather such data.

Keywords- Sports Analytics; Football Analytics; Web-Scrapping; JavaScript; PhantomJS

I. INTRODUCTION

The sports industry has seen a lucrative rise in stature and has now become an important contributor to the global economy. A huge amount of finance is being invested, resulting in an exponential rise in the amount of data generated by sports. This volume of data, together with the development in sports technology, has resulted in the need for informed decisions to be made, more than ever before.

Since Michael Lewis' book "Moneyball" has been published, highlighting the method employed by the GM of Oakland Athletics (American baseball Club) in which he used statistical analysis to build a team on a low budget [1]. Sports Analytics has become a reality and requirement.

Research on Sports Analytics that incorporated both Sports and Software Engineering has flourished over the last few years. Franchises involved in games such as cricket, football, basketball etc. have required performance analysts and statisticians to help understand the game better.

The premise of this case study is the data with which work was done on the project. The idea behind this case study, or project, is hereby explained:

When the idea of working with sports data was formulated, focus was decided to be placed upon association football (hereafter referred to as football) in particular. Thereafter, the first course of action was to search for available online resources in the form of published literature and data. Dataset made publicly available was found by OPTA (in Dutch, Onafhankelijke Post en Telecommunicatie Autoriteit), which is a sports data company which stores and distributes live sports data for over 60,000 fixtures across 30 sports in 70 countries. It is in partnership with Manchester City Football Club (MCFC) in 2012 as part of the MCFC analytics initiative. The data was studied, albeit in an excel sheet, for examination of the information it contained; the way it was structured; the events it recorded, and what each of these meant.

The literary part of the project was then proceeded with, whereby research was done into sports analytics, particularly football analytics. An understanding was gained of past research that had been conducted. It was realized that there was an abundance of aspects of the game that had not been researched upon, and hence could form the main focus of the project. However, the choice of the novel research that may be conducted would be determined by the nature of data at hand.

This case study can be best understood to be a precursor to the eventual methodology selected to develop a project capable of working on Player performance data and extracting valuable insights using predictive modeling and machine learning.

The rest of the study is organized as follows. Section III explains the dataset that is being worked upon, with the details of its release and what information it contains. Section IV is the Literature Survey that was conducted for the eventual

solutions to be followed and worked upon on the project. Section V outlines the problems that were encountered during the course of this study. Section VI presents a walkthrough of the eventual methodology used to solve the problems and gather the rating data required to proceed further with the project.

II. LITERATURE REVIEW

In order to prepare for this project it was vital to read up and study the development that has already been done in this field of research or is being pursued. On recommendation of our internal advisor we began researching and reading articles and research papers that would eventually form part of our literature survey.

Ref. [2] discuss about the development of a player performance rating index and its usage [2].

Ref. [3] provides a vital reading and research source for the possibilities that we could think off and pursue with sports analytics and also gave a good understanding of the technical side of research and development.

The investigation in [4] predicts match outcome based on individual player performance. They use statistical methods such as Bayesian networks, Markov chain Monte Carlo method and Gibbs Sampling.

The study in [5] predicts the premier league season of 2011-2012. They focused on what actually wins soccer matches made use of betting odds and other machine learning techniques to perform predictions.

The work in [6] employs the linear modeling techniques to make predictions on value of soccer players based on their player attributes. This study outlines methodology and methods used for analysis and predictions [6].

III. AIM OF THE PRESENT STUDY

The main objective is to develop a player rating system. A system capable of calculating player rating based on the values of the events recorded in the dataset. From the dataset, attributes were extracted that satisfy a weighted index equation resulting in a rating for each player [2]. With the accumulation of player performance ratings, it would be possible to apply machine learning algorithms for predictive analysis on the data drawing insightful results on player performance. The dataset however, did not contain a rating variable. Hence, extensive data of player by player match by match performance was available, but without a target variable for analysis.

One of the first attempts at calculating ratings was to model the rating system based on goals or shots, the shots could be statistically modeled. Essentially, it was aimed to model match contributions of player as events leading to an outcome that is goals. Thereby, the opposing team earns goals

as a result of their in game contributions. With this possible model it was sought to understand techniques to make use of the 210 attributes at hand to develop a rating equation. Dimensionality reduction was performed. Dimensionality reduction can be defined as applying data encoding and transformation techniques to obtain a reduced version of the original data [7]. The first attempt was to create a single equation that would be applicable on all players. The results were however, un-interpretable. The next move was to apply dimensionality reduction on the data by partitioning the dataset based on the positions the player played that is, creating separate dataset for goalkeepers, defenders, midfielders, strikers. The results again were un-interpretable because the data contained many instances of certain events that were not applicable to players of certain positions and some attributes that could be considered as having only two possible values 1 in case of occurrence within game or 0 otherwise.

IV. DATASET

The Dataset forms the basis of all possible solutions that can be derived through the system; therefore an elaborate and quality dataset is required to assure better model development leading to justified findings. On 17th August 2012 OPTA publicly released a rich dataset of player performance statistics of English Premier League season 2011-2012. It contains frequency counts for hundreds of different events, ranging from aerial duels to yellow cards, broken down by player and by game. This dataset contains close to 10369 rows of data and approximately 210 attributes.

V. METHODOLOGY

At this impasse it was resorted to going back to the research that had already been conducted and commence reading the literature relevant to the available dataset, whereby a method was come across, which was mentioned by [2] where the rating data was gathered from whoscored.com. Whoscored.com is a popular soccer statistic website that uses the same data provided by OPTA to calculate ratings and statistics. Therefore their ratings were tailor-made for the available dataset. The author stated that the data was fetched from whoscored.com and added to the existing dataset to create a new dataset [3].

Our next course of action was to understand the how the whoscored.com website worked and if there were any API's available to leverage their rating data for our use but this did not materialize because there was no such API. The alternative and the eventual method used were to gather the rating data was to create a Web Scraper.

The Document Object Model (DOM) was observed and it was observed that it was, in fact, not a static website but displayed data through AJAX requests. The heavy use of Javascript in displaying match data made it very hard to scrape unless we could decipher a link to the JSON data. Conventional methods would have resulted in our IP address

being blocked by the website because of safeguards to protect from undesirable activity such as Distributed denial-of-service (DDOS) attacks. Therefore the scrapping process had to be automated. There was a need to create an environment that would mimic a user access the website through a browser. This is when PhantomJS was come across, a truly headless Webkit browser that runs JavaScript code, and has no dependencies. With the browser part complete, there was one important aspect to take care of that is the navigation to the page that contained the ratings data required would eventually lead to a callback spaghetti code also known as Callback Hell within the web community. To eradicate this problem, CasperJS was implemented which is a project that is build on top of PhantomJS. Hence forth, scrapping could be commenced whereby, the first task was to gather all the links to the individual matches, from these links it was required to scrap and gather the links to 'player statistics' page, resulting in only having to scrap for the player ratings data and save the results. One important thing to note was that some of the navigation urls, specifically, that of the individual match statistics did not follow any convention therefore, there was quite a task at hand to ensure data was gathered for all the 380 matches.

Once the complete data was gathered it had to be converted from the JSON format and incorporates it into the dataset which was a csv file. This is the next test that was encountered. As mentioned earlier, whoscored.com used the data provided by OPTA there we some differences particularly with in the names of players and teams which became apparent when the dataset created with data from whoscored and the dataset were merged (joined). Keys needed to be created to match each row of the dataset created from whoscored.com and the dataset from OPTA therefore a combined key was created based on the player name, the home team and the away team.

The consistency and of the data needed to be checked; it was noticed that some players represented by their nicknames in either of the two sources from where the data was collected example (Javier Hernandez was named as Chicharito) and some teams with larger names were represented by their shortened names (Wolverhampton Wanderers as Wolves). These discrepancies called for the data to be cleaned. Therefore the reference was set to be the names in the OPTA dataset and transformed the discrepancies in the whoscored.com dataset to match the ones in the OPTA dataset.

VI. CONCLUSION

Through the course of the development of this study we observe that although there is abundance of data on football to

perform operations of varying degree, the nature and the detail that data covers vary from source to source. Information and data relating to the result and outcome of match including betting data is readily available, however when it comes to working on player performance evaluation and player in-game events statistics there is a lack of publicly available data and the presence of high-end data collection and analytics companies working in the field of football analytics developing their own datasets for use by other interested parties lead to a complicated situation. The method of data collection used in this case study outlines the combination of a common trick that is used for gathering data from websites that do not provide access through API's and applies a customized solution of gathering the desired data. The collected data from whoscored.com acts as the second data source therefore the two datasets needed to be combined, it is here that some common ground between datasets was needed which led to the need for data integration and cleaning operation to be performed.

REFERENCES

- [1] M. Lewis, *Moneyball: The Art of Winning an Unfair Game*. W.W.Norton & Company Inc., 2003.
- [2] I. McHale, P. Scarf, and D. Folker, "Soccer Player Performance Rating System," *Articles in Advance*, vol. 42, No. 4, pp.339-351, July–August 2012.
- [3] G. Kumar, 'Machine Learning for Soccer Analytics', Thesis submitted for the degree of Master of Science in Artificial Intelligence, option Engineering and Computer Science, 2013.
https://www.researchgate.net/profile/Gunjan_Kumar2/publication/257048220_Machine_Learning_for_Soccer_Analytics/links/0c96052441dfabfc87000000.pdf
- [4] J. Brigitte and M. Lokketangen, 'Predicting Outcomes of Association Football Matches Based on Individual Players' Performance', Thesis submitted for the Master of Science in Computer Science, Norwegian University of Science and Technology, Trondheim, Department of Computer and Information Science. June 2014. <http://www.diva-portal.org/smash/get/diva2:751709/FULLTEXT01.pdf>
- [5] J. Alan, L. Snyder, 'What Actually Wins Soccer Matches: Prediction of the 2011-2012 Premier League for Fun and Profit' , May 6,2013. <https://homes.cs.washington.edu/~jasnyder/papers/thesis.pdf>
- [6] Yuan He, 'Predicting Market Value of Soccer Players Using Linear Modeling Techniques'.
https://www.stat.berkeley.edu/~aldous/Research/Ugrad/Yuan_He.pdf
- [7] J. Han and M. Kamber, *Data Mining Concepts and Techniques*, 2nd ed. San Francisco: Morgan Kaufmann, 2006, p. 77